Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

1. Recall the formula: Power (P) = Voltage (V) * Current (I) * Resistance (R)

The given power (P) is 40 watts, voltage (V) is 12 volts. The formula can be rearranged to find current (I): I = P / V

I = 40 watts / 12 volts = 3.333 amps

Now, to find the resistance (R), use the formula: R = V / I

R = 12 volts / 3.333 amps = 3.6 ohms

Therefore, the current is 3.333 amps and the resistance is 3.6 ohms

</think><answer>E